METRIC

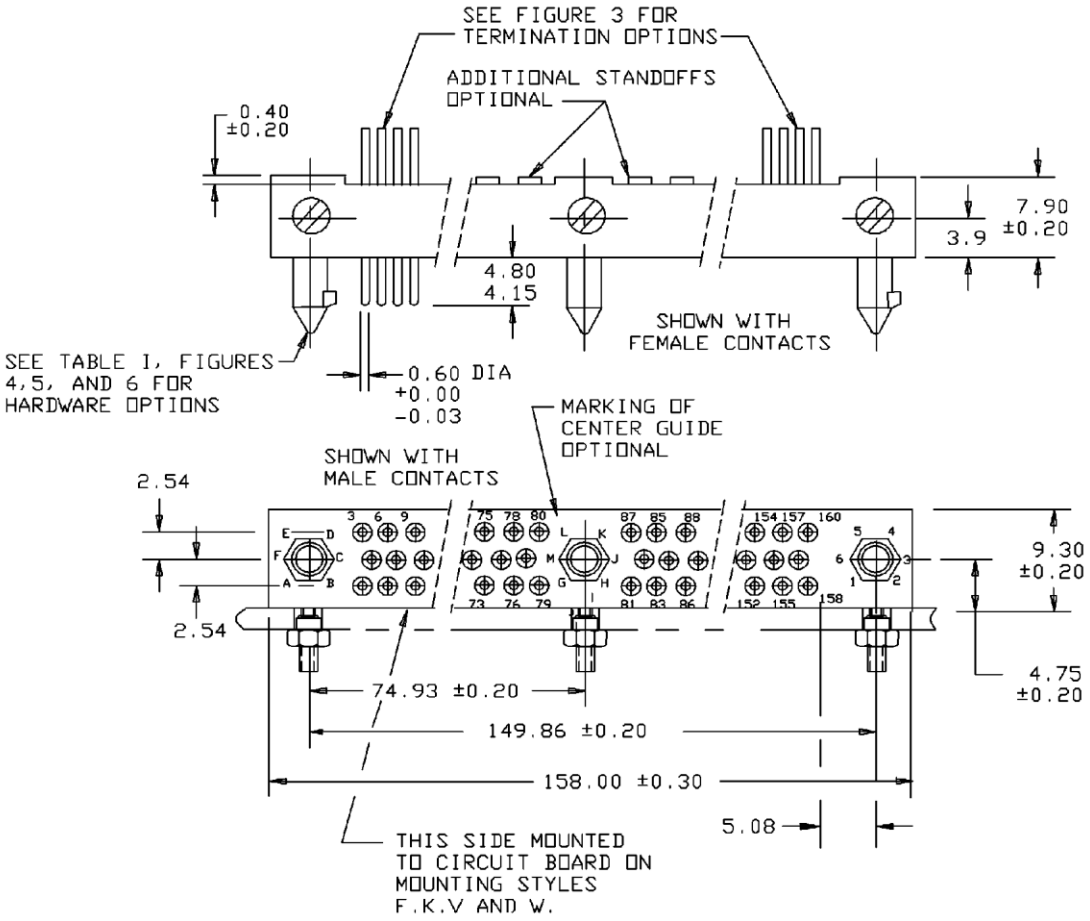
MIL-DTL-55302/164D
w/AMENDMENT 1
20 July 2011
SUPERSEDING
MIL-DTL-55302/164D
8 July 2004

DETAIL SPECIFICATION SHEET

# CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES, PLUG AND/OR RECEPTACLE, 160 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS, 2.54 MM X 1.27 MM OFFSET GRID, METRIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-55302.




NOTE: Dimensions are in millimeters.

FIGURE 1. Connector plug.

AMSC N/A

FSC 5935

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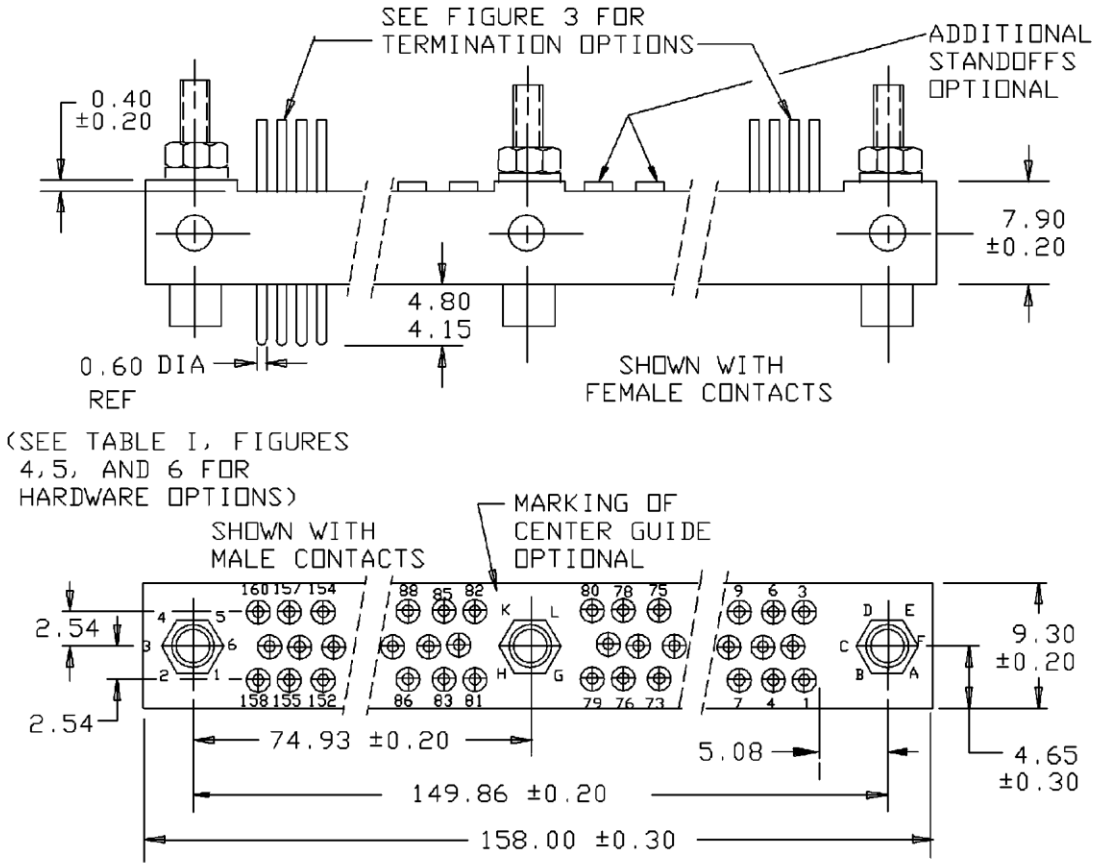



FIGURE 2. Connector receptacle.



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TERMINAL STYLES

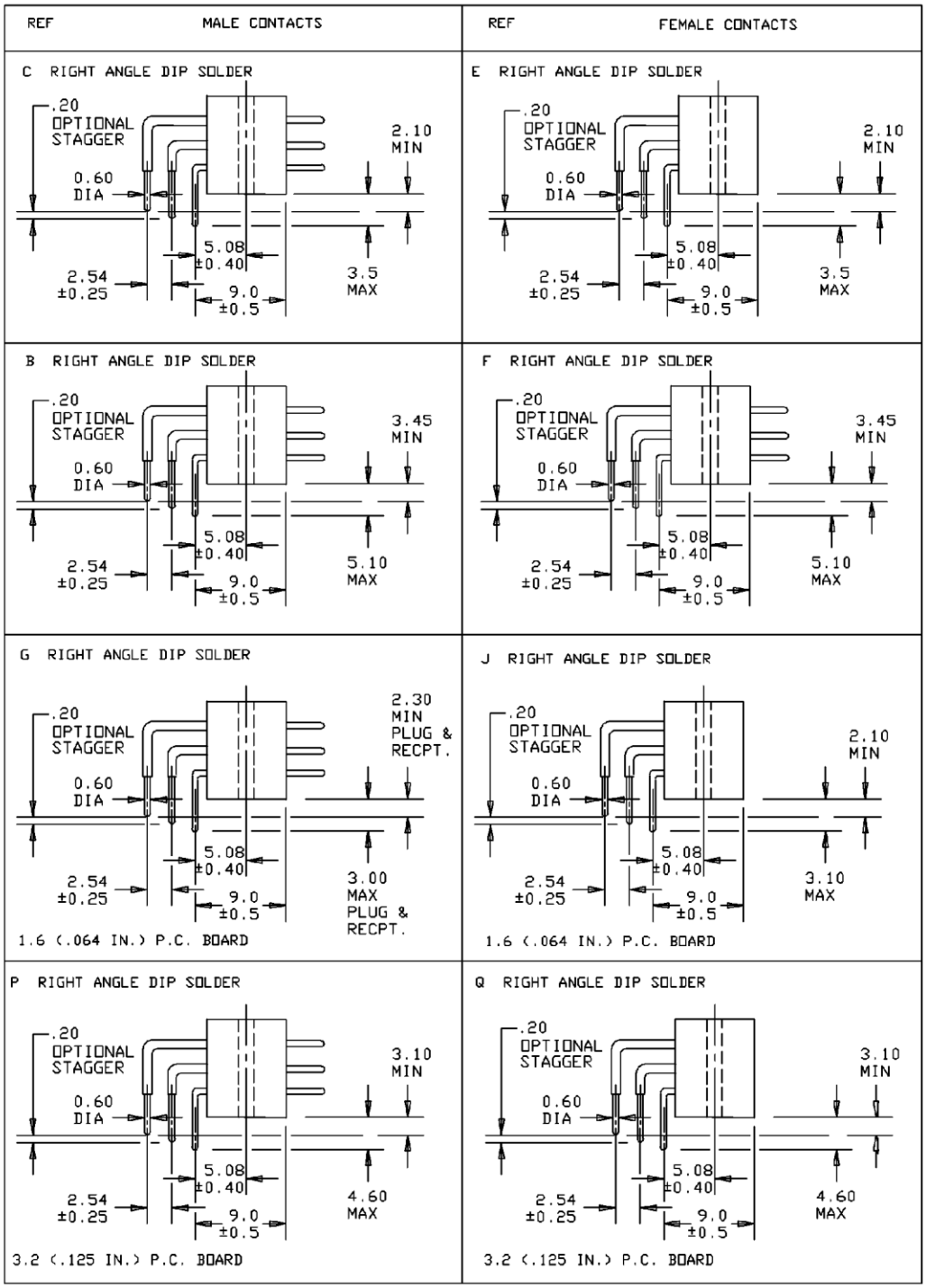



FIGURE 3. Connector termination styles.



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TERMINAL STYLES

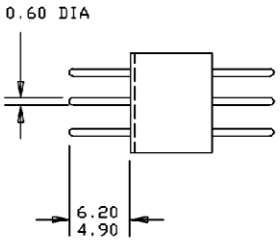
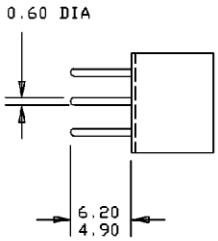
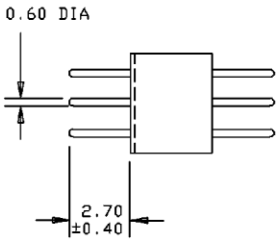
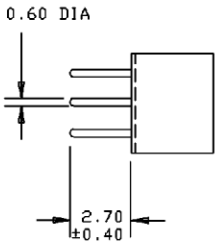
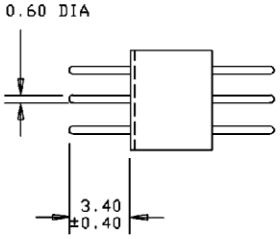
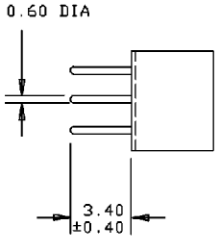
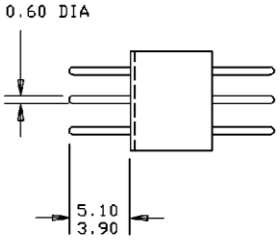
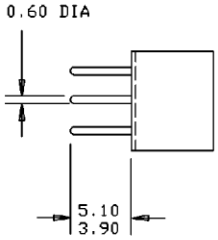
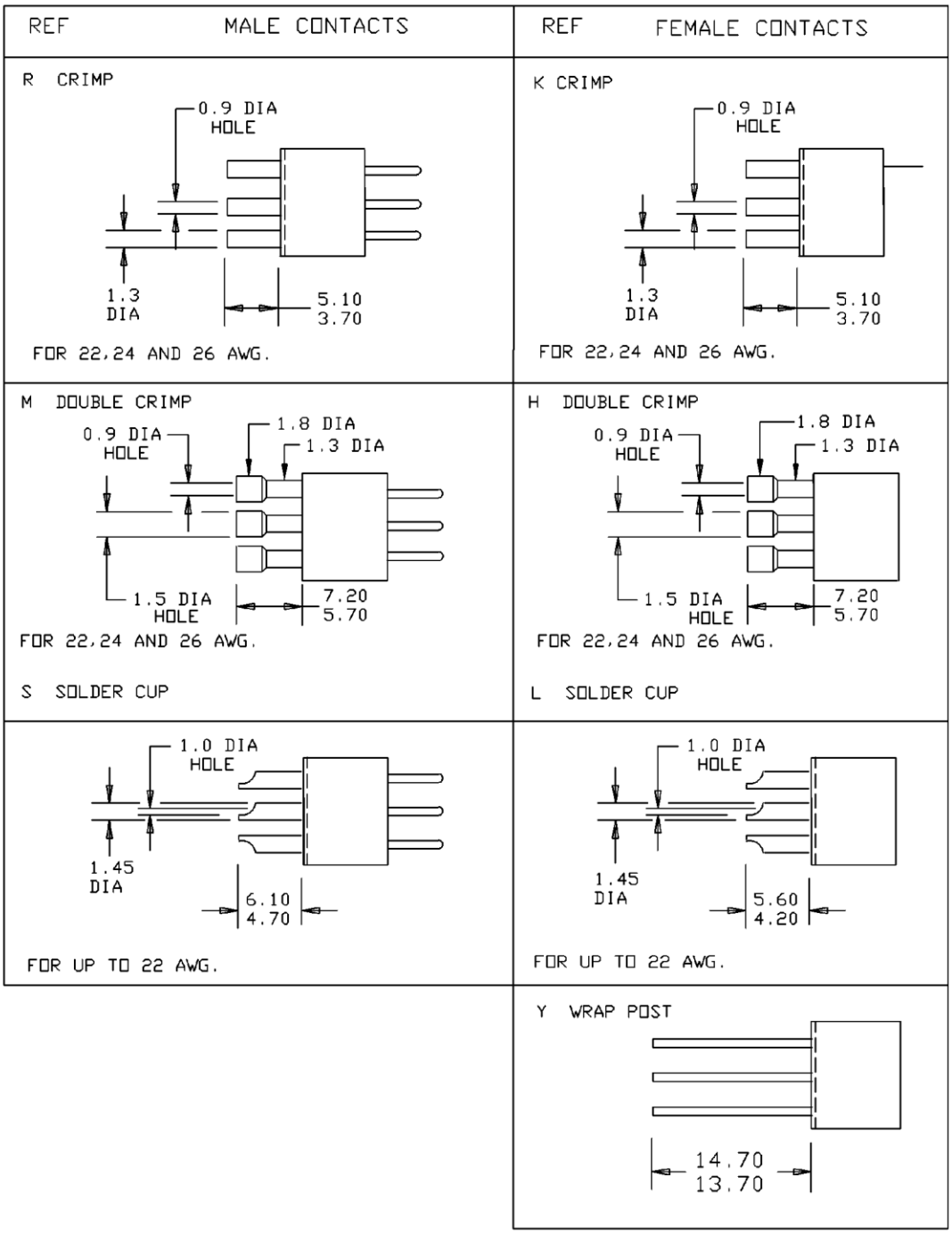
| REF MALE CONTACTS | REF FEMALE CONTACTS |
|---|---|
| A STRAIGHT DIP SOLDER<br>  | D STRAIGHT DIP SOLDER<br>  |
| T STRAIGHT DIP SOLDER<br>  <br>1.6 (.064 IN.) P.C. BOARD | U STRAIGHT DIP SOLDER<br>  <br>1.6 (.064 IN.) P.C. BOARD |
| N STRAIGHT DIP SOLDER<br>  <br>2.4 (.094 IN.) P.C. BOARD | Z STRAIGHT DIP SOLDER<br>  <br>2.4 (.094 IN.) P.C. BOARD |
| V STRAIGHT DIP SOLDER<br>  <br>3.2 (.125 IN.) P.C. BOARD | W STRAIGHT DIP SOLDER<br>  <br>3.2 (.125 IN.) P.C. BOARD |

FIGURE 3. Connector termination styles – Continued.



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TERMINAL STYLES



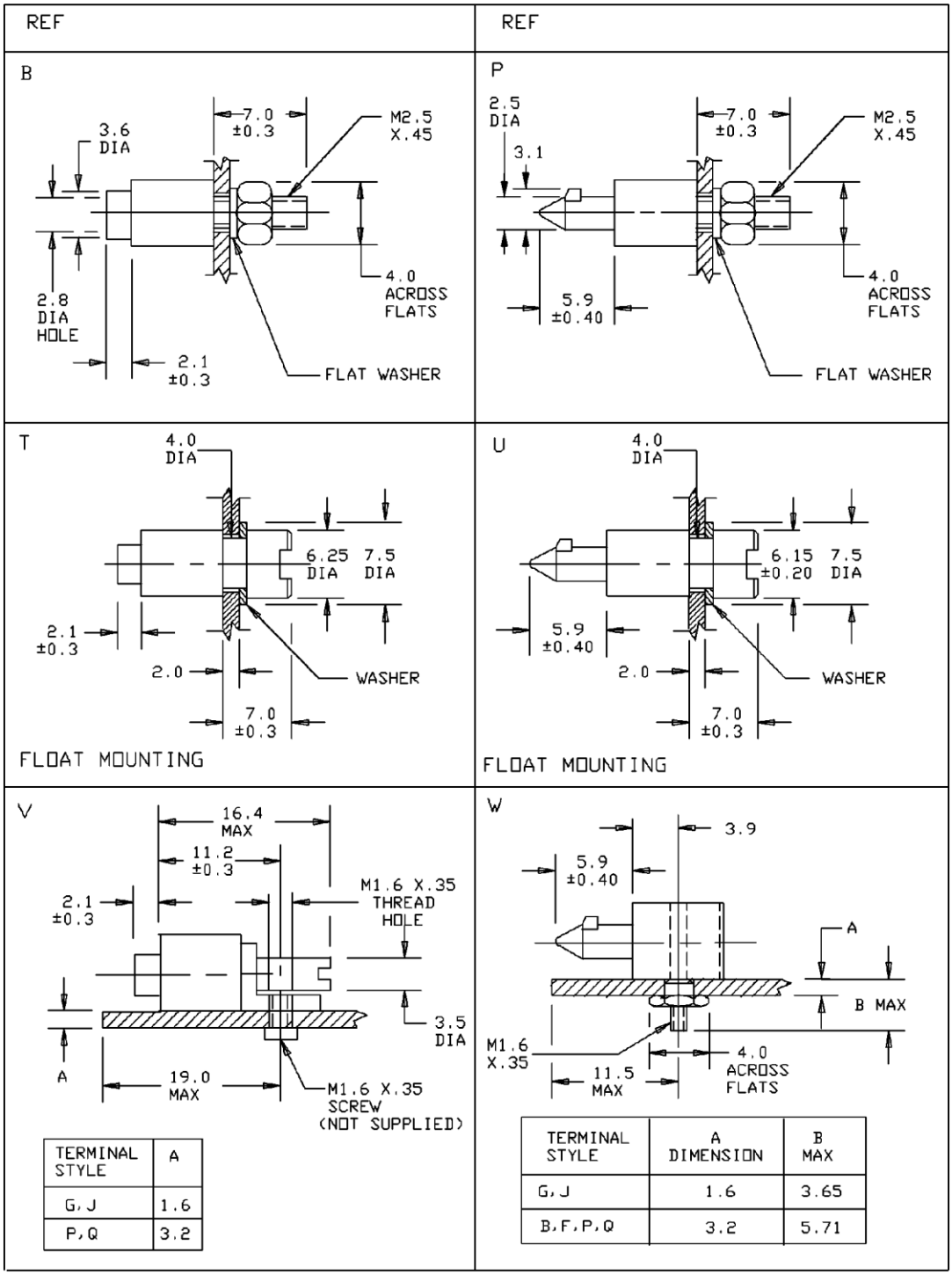

NOTE: Dimensions are in millimeters.

FIGURE 3. Connector termination styles – Continued.



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MOUNTING STYLES




| TERMINAL STYLE | A |
|---|---|
| G, J | 1.6 |
| P, Q | 3.2 |

| TERMINAL STYLE | A DIMENSION | B MAX |
|---|---|---|
| G, J | 1.6 | 3.65 |
| B, F, P, Q | 3.2 | 5.71 |

NOTE: Dimensions are in millimeters.

FIGURE 4. Mounting styles.



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LOCKING HARDWARE

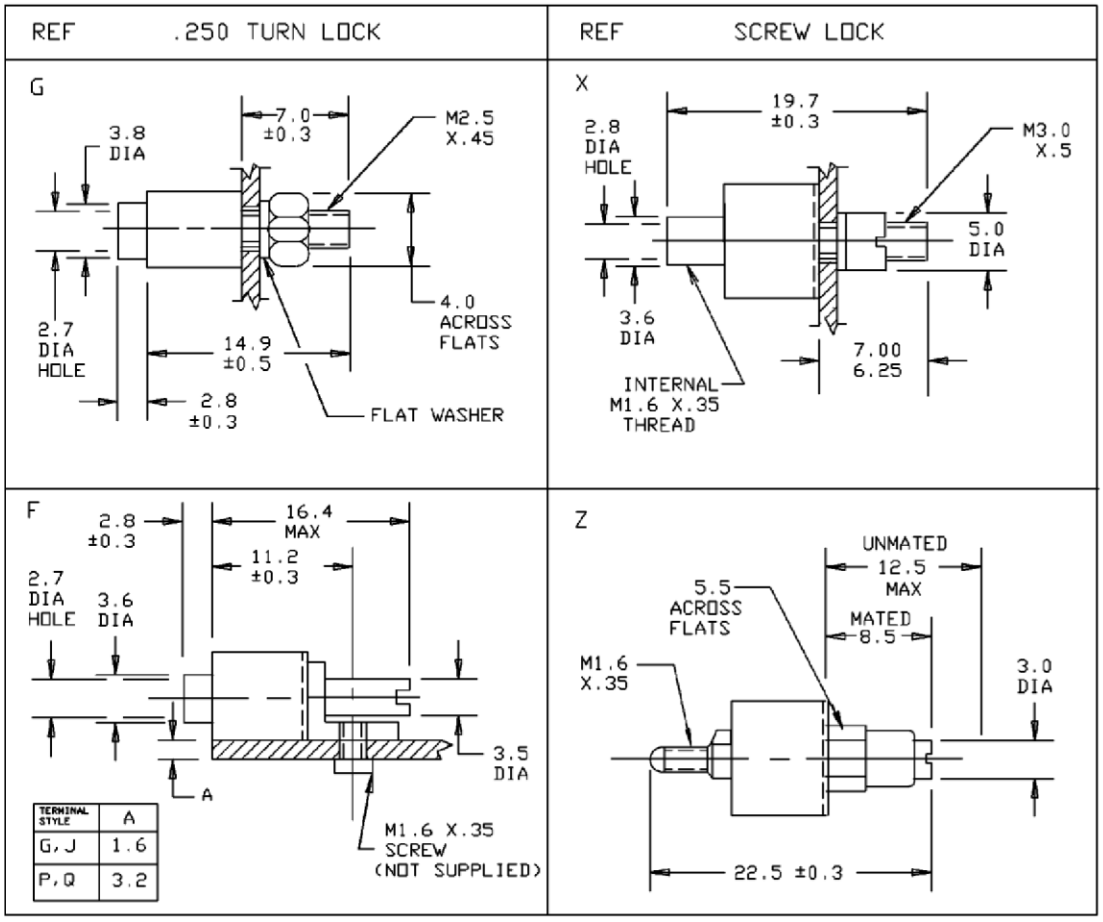
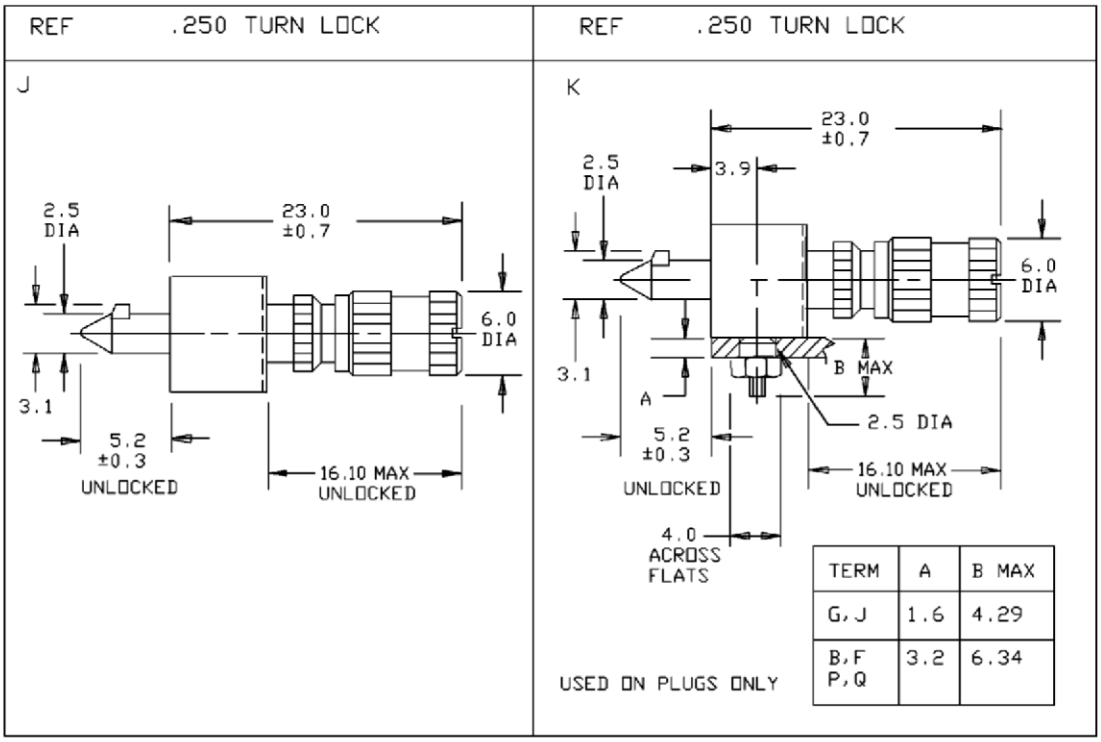



FIGURE 5. Locking hardware.



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LOCKING HARDWARE




| TERM | A | B MAX |
|---|---|---|
| G, J | 1.6 | 4.29 |
| B, F P, Q | 3.2 | 6.34 |

NOTES:
1. Dimensions are in millimeters.
2. This is a metric design.
3. Unless otherwise specified, tolerance is ± 0.10 millimeter.
4. These connectors mate with MIL-DTL-55302/160.
5. See table I for mating.
6. Contact identification may be located adjacent to cavity on face or side and may be rotated 180°.
7. Crimp contacts are shipped not inserted into insulator.
8. Use crimp tool M22520/2-01 for R and K terminal styles with Daniels manufacturing positioner K547.
9. Use crimp tool M22520/2-01 for M and H terminal styles with Daniels manufacturing positioner K547 and K640.

FIGURE 5. Locking hardware – Continued.



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TABLE I. Mating references.

| Type | Will only mate with (see note) | Locking method |
|---|---|---|
| B | P,U,W | None |
| F | J,K | Push 1/4 turn |
| G | J,K | Push 1/4 turn |
| J | F,G | Push 1/4 turn |
| P | B,T,V | None |
| T | P,U,W | None |
| U | B,T,V | None |
| V | P,U,W | None |
| W | B,T,V | None |
| X | Z | Screw |
| Z | X,Y | Screw |

NOTE: These connectors mate with MIL-DTL-55302/160.



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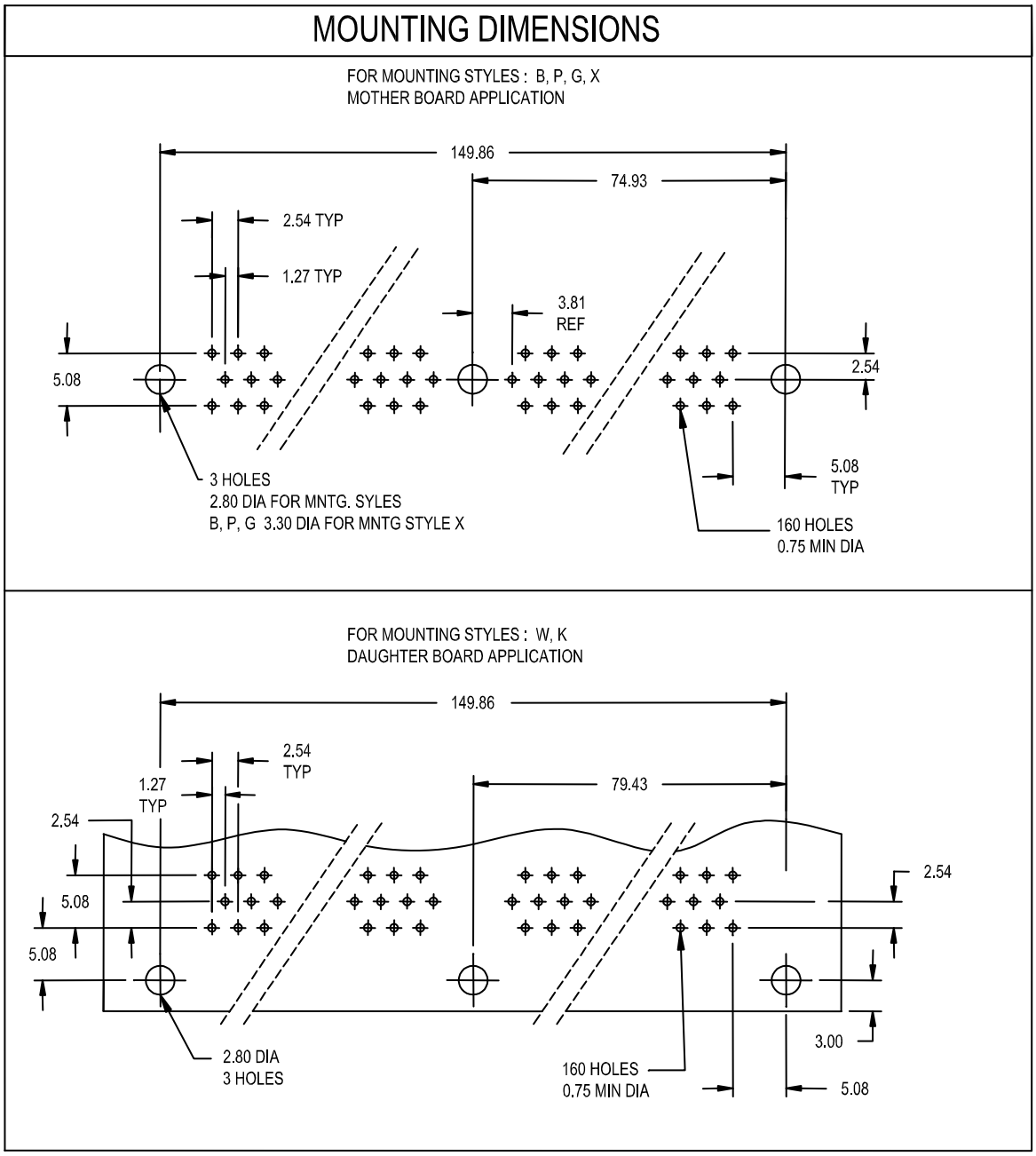



FIGURE 6. Mounting dimensions.



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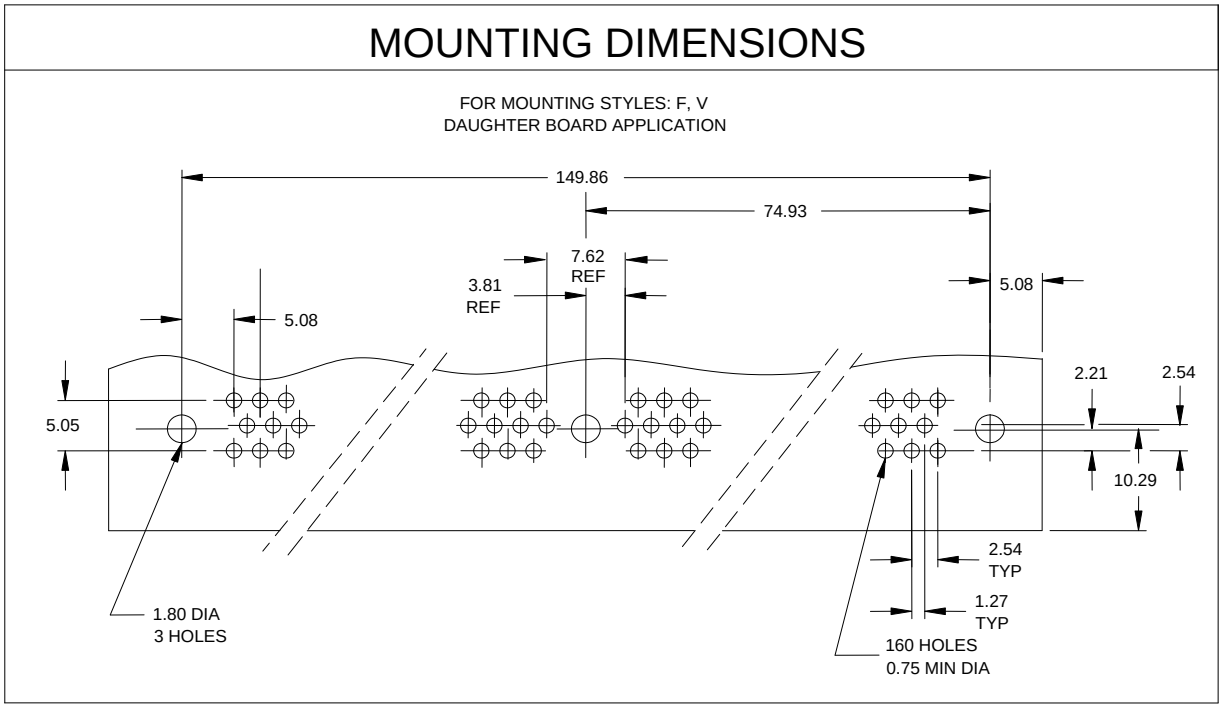
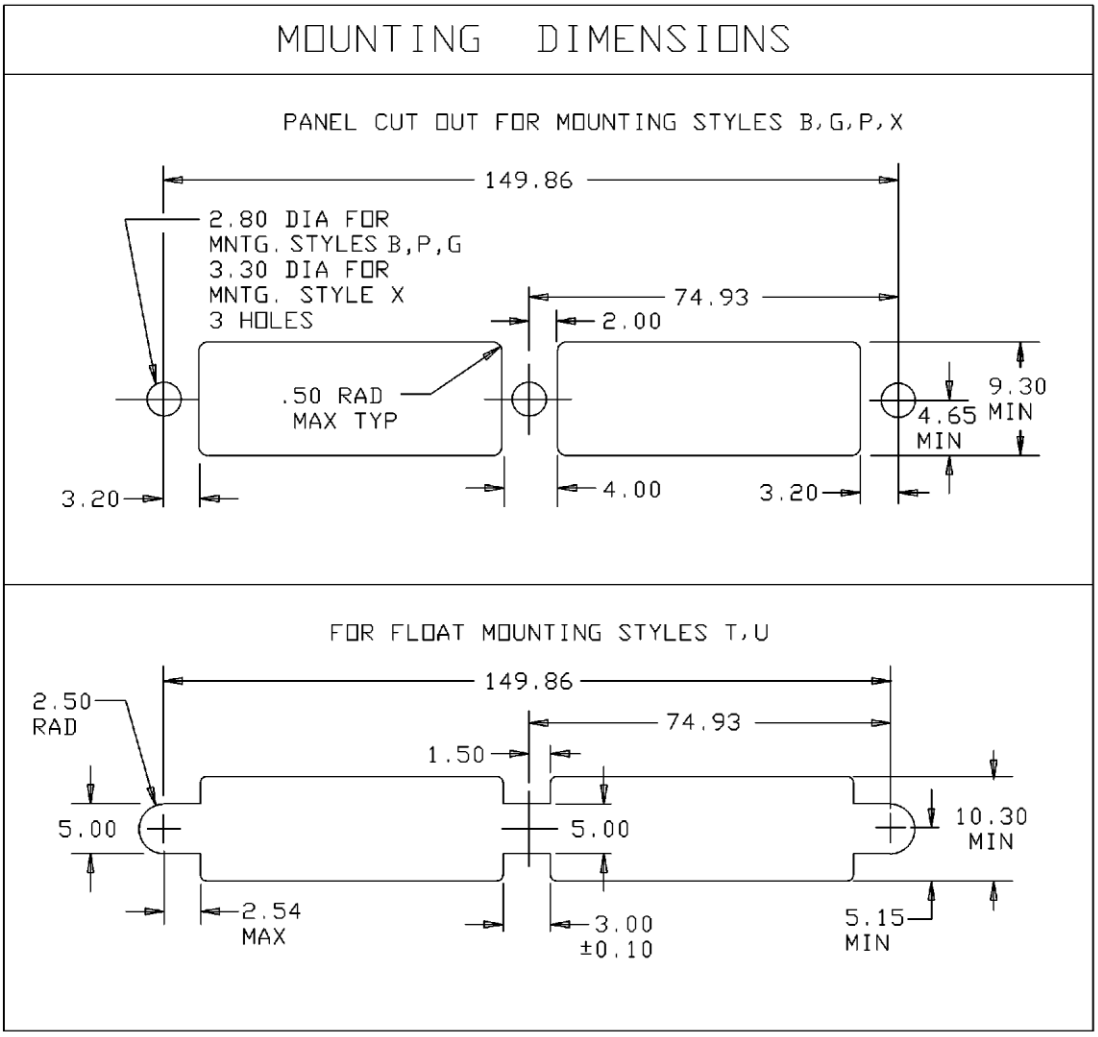



FIGURE 6. Mounting dimensions – Continued.



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NOTES:
1. For plugs only. The printed circuit board may be extended to 11.50 max from center of mounting holes.
2. Tolerances are noncumulative.
3. Dimensions are in millimeters.

FIGURE 6. Mounting dimensions – Continued.



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REQUIREMENTS

Design and construction:

Dimensions and configuration: See figures 1 through 6 and table I.

Material:

Insulators: Diallyl phthalate in accordance with ASTM D5948.

Guides and hardware: Brass in accordance with ASTM B36/B36M, ASTM B16/B16M, ASTM B124/B124M, ASTM B453/B453M, or ASTM B455; nickel plated in accordance with SAE AMS-QQ-N-290 or AISI 300 series stainless steel or equivalent in accordance with ASTM A582/A582M, passivated in accordance with SAE AMS2700.

Plating: Plating in accordance with MIL-DTL-55302 or as noted.

Mating surface (pin and wire): Gold in accordance with ASTM B488, class 1.27, type II, code C over nickel in accordance with SAE AMS-QQ-N-290, 30-150 microinches.

Terminations: Solderable areas (including crimp barrels and wire wrap terminations) shall be gold in accordance with ASTM B488, type III, code A, class 0.254 minimum thickness or gold in accordance with ASTM B488, type II, code C, class 0.51 over nickel in accordance with SAE AMS-QQ-N-290, 30 to 150 microinches thick. Solderable areas must meet the requirements of MIL-STD-202, method 208.

All other surfaces: Gold in accordance with ASTM B488, type II, code C, class 0.127 minimum over nickel in accordance with SAE AMS-QQ-N-290, 30-150 microinches.

Contact identification: See figure 1.

Contact separation forces:

Minimum force: 0.14 newton (0.50 ounce) with a 0.5800 millimeter (.0228 inch) +0.005, -0.00 diameter pin after being cycled 3 times.

Note: Steel pin test surface roughness to be 0.025-0.25 micron rms.

Unmating: The maximum force of a connector assembly shall not exceed 0.53 newton (1.90 ounces) multiplied by the number of contact positions.

Mating: The maximum force of a connector assembly shall not exceed 0.70 newton (2.5 ounces) multiplied by the number of contact positions.

Contact retention minimum: functional value 22.2 newtons (5 pounds); destructive test 35.3 newtons (8 pounds) maximum design and qualification value.

Contact rating: 5 amperes per line.

Durability: 2,000 cycles minimum.



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Dielectric withstanding voltage:

Sea level: 1,400 V ac rms test.

High altitude: 300 V ac rms test.

Keying: Rotation of guides, 6 positions each, 36 possible combinations.

Connector Part or Identifying Number (PIN):

Example: D55302/ 164 D 160 L

Specification number

Specification sheet number

Type of contact and termination
(see figures 1 and 2)

Number of contacts
(see table I)

Guide hardware number
(see figure 3 and table I)

Amendment notations. The margins of this specification are marked with vertical lines to indicate where modifications from this amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-55302, this document references the following:

MIL-DTL-55302/160
MIL-STD-202
ASTM A582/A582M
ASTM B16/B16M
ASTM B36/B36M
ASTM B124/B124M
ASTM B453/B453M
ASTM B455
ASTM B488
ASTM D5948
SAE AMS-QQ-N-290
SAE AMS2700



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CONCLUDING MATERIAL

Custodians:
Army – CR
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2010-131)

Review activities:
Army – AR, AT, AV, CR4, MI

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